Unit 4 Progress Check: MCQ Part A

1. The following SocialMediaAccount class is used to represent a user of a social media site. It includes attributes for the user's username, real name, email address, and ID number.

```
public class SocialMediaAccount
{
  public String username;  // The user's username
  private String realName; // The user's real name
  private String email;    // The user's email address
  private int idNumber;    // The user's ID number

  public String getInfo(String str)
  {
    if (str.equals(username))
    {
      return realName + " " + email + " " + idNumber;
    }
    else
    {
      return "username does not match";
    }
  }

  // Constructor not shown.
  // There are no other methods.
}
```

What is the greatest personal privacy risk posed to users by this class design?

(A) The class contains no mutator methods.

(B) The class utilizes data encapsulation.

(C) Personal information can be determined from outside the class using information that has public visibility. ✓

(D) If the ID numbers are assigned consecutively, the total number of users can be determined.

Answer C

Correct. The instance variable username has public visibility, which means it can be accessed outside the class. If the value of this variable is known, it can be passed to the getInfo method to obtain personal information, including the user's real name and email address.

Unit 4 Progress Check: MCQ Part A

2. The following class is used to store information about a customer.

```
public class Customer
{
  public String cName;        // The customer's name
  public String cPaymentInfo; // The customer's payment information

  /**
   * Returns the customer's name
   */
  public String getName()
  { /* implementation not shown */ }

  /**
   *  Returns the customer's payment information
   *  when a correct password is entered
   */
  public String getPaymentInfo(String password)
  { /* implementation not shown */ }

  /* There may be other methods that are not shown. */
}
```

Which of the following statements best describes the privacy of the customer's payment information?

(A) It is considered private because retrieving it requires a password.

(B) It is considered private because the value of the instance variable `cName` is not easy to guess.

(C) It is not considered private because the instance variables are declared `public`. ✓

(D) It is not considered private because the `getName` method has no password parameter.

Answer C

Correct. Instance variables declared `public` are accessible to any object outside the class.

Unit 4 Progress Check: MCQ Part A

3. A research team is analyzing survey data to determine the average number of hours high school students spend on homework each night. The survey was conducted online, and students volunteered to participate.

Which of the following factors is most important for the team to consider before drawing conclusions from the data?

(A) The survey data are considered reliable because the survey used self-reported responses from students.

(B) The survey results may not represent all high school students because only those with Internet access and interest in the topic participated. ✓

(C) The survey ensures accuracy because students were able to choose whether or not to participate.

(D) The method of survey collection does not matter as long as there are a large number of responses in the data set.

Answer B

Correct. The method of collecting data online and relying on voluntary participation introduces potential biases. These factors can make the sample unrepresentative of the entire population, which is an important consideration when using the data to draw conclusions.

Unit 4 Progress Check: MCQ Part A

4. A retail store has a data set that contains the following information for each sales transaction.

- Store location
- Transaction date
- Time of purchase
- Sales amount, in dollars
- Product category (electronics, clothing, home goods, grocery, etc.)

Which of the following is most easily determined from the data in the data set?

(A) Customer purchasing preferences by category for each store location ✓

(B) Customer satisfaction levels by category for each date

(C) The name of the most popular product at each location

(D) The quantity available for each product at each store location

Answer A

Correct. The time of purchase and product category can be used to analyze the sales for each store location.

Unit 4 Progress Check: MCQ Part A

5. A data set contains information for all high schools in a school district. The following information from the previous year is available for each school in the district.

- The name of the high school
- The number of students who arrived at school late 1 or more times during the year
- The number of students who arrived at school late 10 or more times during the year
- The number of students who submitted medical excuse documentation for 1 or more of their late arrivals
- The percentage of students who voted to delay the school start time by one hour

The following shows a portion of the data set for a district with 13 high schools.

High School Name	Number of Students Late at Least 1 Time	Number of Students Late at Least 10 Times	Number of Students Who Submitted Medical Excuse Documentation	Percent of Student Population Who Voted for Later Start Time
School A	402	59	179	62%
School B	63	2	48	29%
⋮	⋮	⋮	⋮	⋮
School M	163	37	44	84%

Which of the following information **cannot** be determined using only the information in the data set?

(A) The district-wide total number of students who were late to school at least 10 times

(B) The district-wide total number of students who were late to school between 1 and 9 times, inclusive

(C) The district-wide percentage of students who voted for a later school start time ✓

(D) The district-wide percentage of the students who were late at least 1 time who submitted medical excuse documentation for at least 1 late arrival

Unit 4 Progress Check: MCQ Part A

Answer C

Correct. The table does not provide information about the total student population at each school. There is no way to calculate the total student population for the district, nor the total number of students voting to delay start time.

6. A software developer wants to determine which programming language is most commonly used in different types of software projects. They have access to the following data sets.

- Data set 1 contains entries for each registered developer, including their unique developer ID and their favorite programming language.
- Data set 2 contains entries for each software project, including project name, type of project, and the developer ID of the project creator.
- Data set 3 contains entries for each completed project, including project name and user ratings.
- Data set 4 contains entries for each software project, including project name and all programming languages used in the project.

Which two data sets can be combined and analyzed to determine the desired information?

(A) Data sets 1 and 2

(B) Data sets 2 and 4 ✓

(C) Data sets 3 and 4

(D) Data sets 1 and 3

Answer B

Correct. Data set 2 provides information about the type of each project and data set 4 lists all programming languages used in each project. Together they allow analysis of which languages are used across different project types.

Unit 4 Progress Check: MCQ Part A

7. Consider the following code segment.

```
/* missing code */
sequence[1] = 1;
sequence[2] = 2;
```

Which of the following code segments can be used replace /* missing code */ so that the code segment sets the contents of the array to {0, 1, 2}?

(A)

```
int[] sequence;
```

(B)

```
int[] sequence = new int[1];
```

(C)

```
int[] sequence = new int[2];
```

(D) ✓

```
int[] sequence = new int[3];
```

Answer D

Correct. This statement creates an array of length 3. It is initialized with default values {0, 0, 0}. The code segment then modifies the elements at indices 1 and 2, resulting in {0, 1, 2}. The first element (at index 0) remains unchanged from its default value.

Unit 4 Progress Check: MCQ Part A

8. The following code segment is intended to set the contents of the array names to {"Ana", "Billy", "Cathy", "Darius"}. The code segment does not work as intended.

```
String[] names = new String[3]; // Line 1
names[0] = "Ana";               // Line 2
names[1] = "Billy";             // Line 3
names[2] = "Cathy";             // Line 4
names[3] = "Darius";            // Line 5
```

Which of the following changes can be made so that the code segment works as intended?

(A) Changing line 1 to String[] names = new String[4]; ✓

(B) Changing line 1 to String[] names = new names[3];

(C) Changing line 5 to names[2] = "Darius";

(D) Removing line 5

Answer A

Correct. The given code segment creates an array of length 3. Attempting to assign a value to the element at index 3 causes an ArrayIndexOutOfBoundsException because the array is indexed from 0 to 2. By increasing the array length to 4, it allows all four strings to be stored as intended.

Unit 4 Progress Check: MCQ Part A

9. Consider the following code segment.

```
int[] scores = {85, 92, 78, 90, 88};
scores[1] *= 2;            // Line 2
scores[2] += 10;           // Line 3
```

Which of the following best describes the behavior of lines 2 and 3 in the code segment?

(A) Line 2 doubles the first element in the array and line 3 adds 10 to the second element.

(B) Line 2 doubles the second element in the array and line 3 adds 10 to the third element. ✓

(C) Line 2 doubles the third element in the array and line 3 adds 10 to the fourth element.

(D) Lines 2 and 3 do not change the contents of the array.

Answer B

Correct. The statement scores[1] *= 2 doubles the second element (index 1), and the statement scores[2] += 10 adds 10 to the third element (index 2).

Unit 4 Progress Check: MCQ Part A

10. Consider the following code segment.

```
int[] arr = {1, 2, 3, 4, 5};
arr[1] = 4;
arr[3] = 2;
```

Which of the following represents the contents of `arr` after the code segment has been executed?

(A)

{1, 2, 3, 4, 1}

(B)

{1, 3, 3, 1, 5}

(C) ✓

{1, 4, 3, 2, 5}

(D)

{4, 2, 2, 4, 5}

Answer C

Correct. The statement `arr[1] = 4;` sets the element with index 1 (the second element) to 4. The statement `arr[3] = 2;` sets the element with index 3 (the fourth element) to 2.

Unit 4 Progress Check: MCQ Part A

11. The following code segment is intended to set the array numbers to {10, 20, 30, 40}.

```
int[] numbers = new int[4];
/* missing code */
```

Which of the following can replace /* missing code */ so that the code segment will work as intended?

Unit 4 Progress Check: MCQ Part A

(A)

```
int x = 10;

for (int value : numbers)

{

  value = x;

  x += 10;

}
```

(B)

```
for (int j = 0; j < numbers.length; j++)

{

  numbers[j] = j * 10;

}
```

(C)

```
numbers[0] = 10;

for (int j = 0; j <numbers.length; j++)

{

  numbers[j] = numbers[j] + 10;

}
```

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(D)

```
numbers[3] = 40;

for (int j = numbers.length - 1; j > 0; j--)

{

  numbers[j - 1] = numbers[j] – 10;

}
```

Answer D

Correct. This code segment starts at the last element and sets it to 40. It then traverses the array from right to left, assigning each element a value that is 10 less than the element to its right. The code segment correctly sets numbers to {10, 20, 30, 40}.

Unit 4 Progress Check: MCQ Part A

12. Consider the following code segment.

```
int[] nums = {5, 10, 15, 20, 25, 30};
for (int k = 0; k < nums.length - 1; k += 2)
{
   nums[k] = nums[k + 1];
}
```

What are the contents of nums after executing the code segment?

(A)

{5, 5, 15, 15, 25, 25}

(B)

{10, 10, 20, 20, 25, 30}

(C) ✓

{10, 10, 20, 20, 30, 30}

(D)

{10, 15, 20, 25, 30, 30}

Answer C

Correct. The elements at indices 0, 2, and 4 are assigned the values of the elements at the next index. Since k increments by 2, the values at indices 1, 3, and 5 remain the same.

Unit 4 Progress Check: MCQ Part A

13. Consider the following code segment.

```
int[] arr = {10, 20, 30, 40, 50};
for (int x = 1; x < arr.length - 1; x++)
{
  arr[x + 1] = arr[x] + arr[x + 1];
}
```

Which of the following represents the contents of `arr` after the code segment has been executed?

(A)

{10, 20, 50, 90, 50}

(B) ✓

{10, 20, 50, 90, 140}

(C)

{10, 30, 60, 100, 50}

(D)

{10, 30, 60, 100, 150}

Answer B

Correct. The code starts with the second element of the array (i.e., the element at index 1) and adds the current element to the next element of the array. The code will set the third element to `20 + 30`, which is `50`. It will then set the fourth element to `50 + 40`, which is `90`. Finally, it will set the last element to `90 + 50`, which is `140`.

Unit 4 Progress Check: MCQ Part A

14. The Fibonacci numbers are a sequence of numbers. The first two numbers are 1 and 1. Each subsequent number is equal to the sum of the previous two numbers. For example, the first seven Fibonacci numbers are 1, 1, 2, 3, 5, 8, and 13.

The following code segment is intended to fill the `fibs` array with the first ten Fibonacci numbers. The code segment does not work as intended.

```
int[] fibs = new int[10];
fibs[0] = 1;
fibs[1] = 1;
for (int j = 1; j < fibs.length; j++)
{
  fibs[j] = fibs[j - 2] + fibs[j - 1];
}
```

Which of the following best explains why the code segment does not work as intended?

(A) In the `for` loop header, the initial value of `j` should be `0`.

(B) In the `for` loop header, the initial value of `j` should be `2`. ✓

(C) The `for` loop condition should be `j < fibs.length - 1`.

(D) The `for` loop condition should be `j < fibs.length + 1`.

Answer B

Correct. If `j` starts at `1`, the attempt to access `fibs[j - 2]` during the first loop iteration will throw an `ArrayIndexOutOfBoundsException`. Starting the loop at `j = 2` avoids this error.

Unit 4 Progress Check: MCQ Part A

15. Consider the following code segment.

```
int[] arr1 = {2, 4, 6, 8, 10};
int num = 0;
for (int val : arr1)
{
   num += 10;
}
```

Which of the following best describes the behavior of the code segment?

(A) The loop adds 10 to each element in `arr1`.

(B) The loop adds 10 to `num` once for each element in `arr1`. ✓

(C) The loop adds the value of each element in `arr1` to `num`.

(D) The loop does not change the value of `num` because it uses an enhanced `for` loop.

Answer B

Correct. The enhanced `for` loop iterates for each element in `arr1`. For each iteration, the value 10 is added to `num`.

Unit 4 Progress Check: MCQ Part A

16. The following method is intended to return an array with the values of the parameter `letters` in reverse order. The method does not work as intended.

```
public static String[] reverseArray(String[] letters)
{
  String[] revLetters = new String[letters.length];  // Line 3
  for (int x = 0; x < letters.length; x++)           // Line 4
  {
    revLetters[x] = letters[letters.length - x];    // Line 6
  }
  return revLetters;
}
```

Which of the following changes can be made so the method returns the intended value?

(A) Changing line 3 to `String[] revLetters = new String[letters.length - 1];`

(B) Changing line 4 to `for (int x = 0; x < letters.length - 1; x++)`

(C) Changing line 6 to `revLetters[x] = letters[letters.length - x - 1];` ✓

(D) Changing line 6 to `letters[x] = revLetters[letters.length - x];`

Answer C

Correct. The original line of code would be off by one and cause an `ArrayIndexOutofBoundsException` when attempting to access the element at index `letters.length`. Subtracting 1 from the index will ensure that each element of `revLetters` is assigned an element of `letters`. As `revLetters` is traversed from left to right, it is assigned a value from `letters` from right to left.

Unit 4 Progress Check: MCQ Part A

17. In the following code segment, data is an int array with at least two elements.

```
int answer = 0;
for (int val : data)
{
   if (data[0] < val)
   {
      answer++;
   }
}
System.out.println(answer);
```

Which of the following best describes the value printed by the code segment?

(A) The number of negative elements in data

(B) The number of positive elements in data

(C) The number of elements in data that are greater than the first element in data ✓

(D) The number of elements in data that are less than than the first element in data

Answer C

Correct. The enhanced for loop compares each array value with the first element of the array (data[0]) and counts the values larger than the first element.

Unit 4 Progress Check: MCQ Part A

18. In the following code segment, assume that the int array numArr has been properly declared and initialized. The code segment is intended to reverse the order of the elements in numArr. For example, if numArr initially contains {1, 3, 5, 7, 9}, it should contain {9, 7, 5, 3, 1} after the code segment executes.

```
/* missing loop header */
{
  int temp = numArr[k];
  numArr[k] = numArr[numArr.length - k - 1];
  numArr[numArr.length - k - 1] = temp;
}
```

Which of the following can be used to replace /* missing loop header */ so that the code segment works as intended?

(A) ✓

```
for (int k = 0; k < numArr.length / 2; k++)
```

(B)

```
for (int k = 0; k < numArr.length; k++)
```

(C)

```
for (int k = 0; k < numArr.length / 2; k--)
```

(D)

```
for (int k = numArr.length - 1; k >= 0; k--)
```

Answer A

Correct. With this loop header, the code segment will iterate through the array from left to right, swapping the element k positions from the front of the array with the element k positions from the end of the array. The loop terminates when k has reached the middle of the array, leaving the array elements in reverse order.